

MEAM 517 Problem Set 1

Due Thursday September 3, 2026

1 Discrete-time System with Feedback Delay

For a discrete linear system, $x_{t+1} = Ax_t + Bu_t$, with a linear feedback control law $u_t = Kx_t$, the closed-loop system is stable if and only if $\rho(A + BK) < 1$, where $\rho(M) = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } M\}$ and $|\lambda|$ denotes the complex modulus.

- a) Consider the case where feedback control is delayed by a time step such that $u_t = Kx_{t-1}$. In this case, what condition is both necessary and sufficient for the stability of the system? (Hint: define a new “state” as $z_t = [x_t, x_{t-1}]^T$)
- b) What about a feedback law with a delay of two time steps ($u_t = Kx_{t-2}$)?
- c) Now consider the linear discrete system:

$$x_{t+1} = \begin{bmatrix} 0.5 & 0 \\ 0 & 1 \end{bmatrix} x_t + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_t \quad (1)$$

Find a K such that under the control law $u_t = Kx_t$, the system is stable, but under the control law $u_t = Kx_{t-1}$ the system is unstable?

2 Uncertain Pendulum

Consider an inverted pendulum with damping, which you can approximately describe by the linear system:

$$\dot{x} = \begin{bmatrix} \dot{\theta} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \omega^2 & -\gamma \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u \quad (2)$$

for some $\omega^2 > 0$ and $\gamma \geq 0$. If u follows a linear feedback controller, $u = -[k_1 \ k_2] x_t$, we want to determine what gains k_1, k_2 will stabilize the system. Unfortunately, we don't know ω^2 and γ exactly, but only that they lie in the ranges $\omega^2 \in [5, 10]$ and $\gamma \in [0, 1]$. Describe the set of all (k_1, k_2) that stabilize ALL potential values of ω^2 and γ .

3 Exponential Stability and Friction

Consider an actuated 1D block with friction governed by the system:

$$\ddot{x} = \begin{cases} 0, & \dot{x} = 0 \text{ and } |u| \leq \mu & (\textit{sticking}) \\ u - \mu, & \dot{x} > 0 \text{ or } (\dot{x} = 0 \text{ and } u > \mu) & (\textit{sliding}) \\ u + \mu, & \text{otherwise} \end{cases} \quad (3)$$

where $\mu > 0$ is the coefficient of friction. Prove that there are no gains k_p, k_d such that the control law

$$u = -k_p x - k_d \dot{x} \quad (4)$$

is exponentially stable around the origin $(0, 0)$.

4 Gradient Flow

Consider performing *gradient flow* (continuous version of gradient descent) on a function $l : \mathbb{R}^n \rightarrow \mathbb{R}$. This results in the dynamical system:

$$\dot{x} = -\nabla l(x) \quad (5)$$

Additionally, a twice-differentiable function l is considered *strongly convex* if, for some $\sigma > 0$ and all $x \in \mathbb{R}^n$, its Hessian satisfies: $H - \sigma I$ is positive semi-definite, or equivalently that the minimum eigenvalue of the Hessian is lower-bounded by σ :

- a) If l is strongly convex, does this mean that the dynamical system formed by gradient flow has a stable equilibrium?
- b) Consider the quadratic function $l(x) = x^\top Q x$ where $Q \succeq \sigma I$ for some $\sigma > 0$. Gradient flow can be discretized into gradient descent for a step size $\Delta t > 0$, into the system:

$$x_{t+1} = x_t - \Delta t \nabla l(x_t) \quad (6)$$

Find the set of all $\Delta t > 0$ such that the discrete-time dynamical system is globally stable for our quadratic l , in terms of σ .

5 2-D Equilibrium

Consider the following system with state $x = [x_1; x_2]$:

$$\dot{x} = f(x) = \begin{bmatrix} x_2 - 2 \\ -x_1 - x_2 + \frac{1}{3}(x_1 + 1)^3 + 1 \end{bmatrix} \quad (7)$$

- (a) Find all equilibria of the system; that is, find all x such that $f(x) = 0$.

- (b) Determine the single stable equilibrium x^* of the system, and show that it is stable by analyzing appropriately linearized dynamics.
(Hint: you can define a change of coordinates $z = x - x^*$, and linearize the shifted dynamics $\dot{z} = f(z + x^*)$ at the equilibrium $z^* = 0$).
- (c) In Python, construct a 2-D phase portrait of the system over a domain of height and width 8 with the stable equilibrium centered. You may find the functions `numpy.quiver(X, Y, U, V)` and `numpy.meshgrid(x, y)` useful. Note that the absolute magnitude of the arrows plotted on the portrait is unimportant; `quiver` may resize the arrows for readability.
- (d) Examining your phase portrait, overlay an approximation for the boundary of the region of attraction for x^* (See: `scipy.spatial.ConvexHull()`).
- (e) Using `scipy.integrate.solve_ivp(f, (t0, tf), x0)`, simulate the system for two initial conditions: one inside the region of attraction, such that convergence to x^* is observed; and one outside of the region of attraction. Overlay these trajectories onto your portrait from (c).

6 1-D Phase Plot

Consider the system

$$\dot{x} = x^2 + a \tag{8}$$

for some constant $a \in \mathbb{R}$.

- (a) Graph a 1-D phase plot for (8) with $a = -1$ over a domain including all equilibria.
- (b) Graph a bifurcation diagram of (8) over the domain $-1 < a < 1$; label equilibria as stable or unstable. For what values of a does the system contain a stable equilibrium?
- (c) Assuming that a stable equilibrium exists, derive an expression for its region of attraction in terms of a .

7 Stability of Nonlinear Systems

Consider the nonlinear system where $x = [r; \theta]$ is expressed in polar coordinates $r \geq 0, \theta \in [-\pi, \pi)$:

$$\begin{aligned} \dot{r} &= -r + \theta + 1, \\ \dot{\theta} &= \sin(\theta/2)^2. \end{aligned}$$

- (a) Show that $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is an equilibrium.

- (b) Is the equilibrium locally asymptotically stable? Explain.
- (c) For any initial condition, what is $\lim_{t \rightarrow \infty} x(t)$? Explain.